

Challenge (Habanero)

- Q1.** Hikers need to cover $15 \cdot 3 + 8$ miles to reach their campsite. How many miles will they cover in total?
- (A) 43
(B) 47
(C) 51
(D) 55
(E) 59
- Q2.** A park has x deer and $2x + 5$ rabbits. If the total number of animals is 47, what is the value of x ?
- (A) 6
(B) 7
(C) 8
(D) 9
(E) 10
- Q3.** A car travels 240 miles on 12 gallons of fuel. How many miles does it travel per gallon, expressed as $x \cdot \frac{\text{miles}}{\text{gallon}}$?
- (A) $20 \cdot \frac{\text{miles}}{\text{gallon}}$
(B) $18 \cdot \frac{\text{miles}}{\text{gallon}}$
(C) $15 \cdot \frac{\text{miles}}{\text{gallon}}$
(D) $12 \cdot \frac{\text{miles}}{\text{gallon}}$
(E) $10 \cdot \frac{\text{miles}}{\text{gallon}}$
- Q4.** What is the simplified expression of $3(2x - 1) + 2(x + 4)$?
- (A) $6x + 10$
(B) $8x - 2$
(C) $8x + 2$
(D) $7x + 7$
(E) $5x + 5$
- Q5.** A group of friends want to hike a trail that is $8 \cdot 2 + 11$ kilometers long. How many kilometers is the trail?
- (A) 19
(B) 21
(C) 23
(D) 25
(E) 27
- Q6.** The population of a bird species is modeled by the expression $25 + 5x - 2x^2$. What is the population when $x = 3$?
- (A) 10
(B) 15
(C) 16
(D) 19
(E) 22
- Q7.** A hiker has x liters of water and wants to divide it equally among 4 friends. How many liters will each friend get, expressed as $x \cdot \frac{\text{liters}}{\text{friend}}$?
- (A) $\frac{x}{4} \cdot \frac{\text{liters}}{\text{friend}}$
(B) $\frac{x}{2} \cdot \frac{\text{liters}}{\text{friend}}$
(C) $\frac{3x}{4} \cdot \frac{\text{liters}}{\text{friend}}$
(D) $2x \cdot \frac{\text{liters}}{\text{friend}}$
(E) $x \cdot \frac{\text{liters}}{\text{friend}}$
- Q8.** A vehicle consumes 12 gallons of fuel to travel 360 miles. How many gallons will it consume to travel x miles, expressed as $x \cdot \frac{\text{gallons}}{\text{mile}}$?
- (A) $\frac{x}{30} \cdot \frac{\text{gallons}}{\text{mile}}$
(B) $\frac{x}{25} \cdot \frac{\text{gallons}}{\text{mile}}$
(C) $\frac{x}{20} \cdot \frac{\text{gallons}}{\text{mile}}$
(D) $\frac{x}{15} \cdot \frac{\text{gallons}}{\text{mile}}$
(E) $\frac{x}{10} \cdot \frac{\text{gallons}}{\text{mile}}$

Open Ended Questions (FRQ)

- Q9.** The population of a rabbit species is modeled by the expression $10x^2 + 20x + 5$. Find the population when $x = 4$ and show all your work. You may use the space below for calculations.
- Q10.** A car travels x miles and consumes $x + 5$ gallons of fuel. If the car travels 250 miles, how many gallons of fuel will it consume? Show all your work and explain your reasoning. You may use the space below for calculations.

Chef's Tips

- Make sure to check for the correct order of operations
- Encourage students to simplify expressions and combine like terms
- Remind students to substitute values carefully and evaluate expressions correctly